

sources of noise
shrink with more data

design choices
need a sensitivity check

real differences between items
 σ_α^2

Content
*posts, articles,
open-ended responses*

category sampling,
sample size N

rewording the rubric
changes labels
 σ_ρ^2

annotation layer

Coding rubric
instructions for the LLM coder

rubric wording,
scale anchors

noise from one call to the next σ_ε^2
LLM coders disagree on items
(*inter-coder disagreement*) $\sigma_{\alpha\lambda}^2$

LLM annotation
coder's label for the item

which LLM, temperature,
scoring rule (binary / Likert / BT)

Aggregate estimate $\hat{\theta}_{hm}$